

BINSHUAI (DEREK) WANG

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EDUCATION

George Washington University

Jan. 2022 – Present

Ph.D. candidate in Computer Science

Supervisor: Peng Wei

University of California, Davis

Sep. 2019 – Dec. 2021

M.S. in Computer Science

Beihang University

Sep. 2014 – Jul. 2018

B.S. in Mathematics and Applied Mathematics

Hua Luogeng Honor Class

RESEARCH INTERESTS

Optimal Transport, Machine Learning (Vision Transformers, Diffusion Models, and Deep Neural Network Optimization), Convex Optimization, Reinforcement Learning, and Hashing Methods.

RESEARCH EXPERIENCE

George Washington University, Washington, DC

Jan. 2022 – Present

- Developed the Linear Reusable Neural Bases Architecture (LRNBA), a novel RNN framework that represents each network block as a linear combination of shared neural bases, reducing memory costs while allowing for the construction of much wider and deeper models. Experiments on ResNet and Transformer models demonstrate faster convergence, better generalization, and lower final loss, while preserving stable training dynamics and a reduced memory usage.
- Introduced a geometric framework for measuring non-Gaussianity using a relative Wasserstein angle and projection distance, derive closed-form solutions in one dimension, and develop a numerical approximation method for higher dimensions, validated on synthetic and real datasets.
- Proposed a distribution-based evaluation framework to quantify differences between training and real-world data in computer vision tasks, and applied diffusion models to generate synthetic image data for data-scarce cases; received the Best of Session Award at DASC 2025.
- Trained a Vision Transformer to encode sequential thunderstorm snapshots for retrieving similar thunderstorm patterns.
- Developed the relative translation-invariant Wasserstein distance, a novel probability metric with simple and efficient LP-based or Sinkhorn-based implementations, and applied it to large-scale weather-pattern retrieval.

University of California, Davis, Davis, CA

Nov. 2020 – Dec. 2021

- Developed an algorithm for generating high-dimensional cutting planes in integer programming by formulating the task as verifying intersections among pairs of polyhedra and employing an R-tree with adaptive projection techniques to efficiently identify intersecting pairs. The algorithm has been implemented in object-oriented programming (OOP) for Sagemath.

Beihang University, Beijing, China

Sep. 2016 – Jul. 2019

- Proposed a new family of locality-sensitive hashing functions for the problem of subspace-to-subspace search applied in large-scale datasets. Those functions preserve the angular distances. Experiments show that the proposed approach outperforms the state-of-the-art methods, in terms of both accuracy and efficiency (up to $16\times$ speedup); awarded Best Student Paper at PCM 2018.

WORK EXPERIENCE

National Center for Atmospheric Research

Research Intern

Jun. 2023 – Jul. 2023

Boulder, CO

- Developed a probability-based method for identifying similar thunderstorm patterns using optimal transport theory.

PAPERS

- **Binshuai Wang**, Peng Wei. “A Mathematical Theory of Reusable Neural Bases for Network Compression,” arXiv preprint arXiv:2609.01550, 2026.
- Mahyar Ghazanfari, Amin Tabrizian, Arsyi Aziz, **Binshuai Wang**, Peng Wei. “A Paragraph is Worth a Thousand Captions: Rethinking Text Supervision for Vision-Language Retrieval”, European Conference on Computer Vision (ECCV), 2026.
- **Binshuai Wang**, Peng Wei. “Relative Wasserstein Angle and the Problem of the W_2 -Nearest Gaussian Distribution”, arXiv preprint arXiv:2601.22355, 2026.
- **Binshuai Wang**, Qiwei Di, Ming Yin, Mengdi Wang, Quanquan Gu, Peng Wei. “Relative-Translation Invariant Wasserstein Distance”, Transactions on Machine Learning Research (TMLR), 2026.
- **Binshuai Wang**, Can Vu, Peng Wei. “Performance Estimation of Aircraft Computer Vision Models using Optimal Transport-based Distributional Mismatch”, **Best of Session Award**, IEEE Digital Avionics Systems Conference, 2025.
- **Binshuai Wang**, Peng Wei. “Thunderstorm Pattern Analysis Using Vision Transformers and Optimal Transport for Aviation Operations”, AIAA AVIATION, 2025.
- Amber Liu, **Binshuai Wang**, Peng Wei. “Identifying Similar Weather Patterns Using Modified p-Wasserstein Distances”, AIAA SciTech, 2024.
- **Binshuai Wang**, Peng Wei. “Identifying Similar Thunderstorm Sequences for Airline Decision Support via Optimal Transport Theory”, AIAA SciTech, 2023.
- Yi Xu, Xianglong Liu, **Binshuai Wang**, Ke Xia, Xianbin Cao. “Fast Nearest Subspace Search via Random Angular Hashing”, IEEE Transactions on Multimedia, vol. 23, pp. 342-352, 2021.
- **Binshuai Wang**, Xianglong Liu, Ke Xia, Kotagiri Ramamohanarao, Dacheng Tao. “Random Angular Projection for Fast Nearest Subspace Search”, **Best Student paper**, 19th Pacific-Rim Conference on Multimedia (PCM), 2018.

SKILLS

Languages: Python, C++, CUDA, Pascal, HTML, JavaScript, SQL.

Systems/Platforms: Unix/Linux, Google Cloud Platform, AWS, Azure.

Tools: Git, PyTorch, TensorFlow, PySpark, Pandas, Mathematica, MATLAB, Gurobi, CPLEX, MongoDB, React, Bluesky, Ray.

HONORS AND AWARDS

Best of Session Award, IEEE Digital Avionics Systems Conference (DASC)	2025
Graduate Research Award, George Washington University	2022-2026
Best Student Paper Award, Pacific-Rim Conference on Multimedia (PCM)	2018
Outstanding Undergraduate Thesis, Beihang University	2018
Honorable Mention, COMAP’s Mathematical Contest in Modeling (MCM)	2017, 2016